Ria S.

Software Developer · Data Science
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PROFILE SUMMARY

Software Engineer and Data Science student with demonstrated experience shipping full-stack, cloud-deployed applications. Adept at integrating ML/AI pipelines, computer vision, and AWS infrastructure into production systems, with a strong interest in low-level systems, and two research papers accepted: EDGEI 2026 (Springer LNNS) and IMPACT 2026 (Springer ICT series), both forthcoming. Passionate about building intelligent, scalable solutions — from wardrobe AI to an x86 operating system written from scratch.

EDUCATION

B.Tech — Computer Science & Data Science 2024 – 2028
Manipal University Jaipur · Jaipur

WORK EXPERIENCE

Data Science & Analytics Intern Oct 2025 – Dec 2025
Future Interns — Remote
- Conducted end-to-end analysis on e-commerce and social media datasets; delivered insights on customer behaviour trends.

Software Engineering Job Simulation Jul 2025
J.P. Morgan · Forage
- Built financial data tooling prototypes; worked with real-world engineering workflows using JPMorgan's Perspective library.

Data Analytics Virtual Experience Jun 2025
Quantium · Forage
- Analysed retail transaction data to uncover purchasing trends and communicate strategic recommendations.

Data Analytics Virtual Internship Jun 2025
Deloitte · Forage
- Prepared, modelled, and visualised forensic data for business storytelling using Tableau and Excel.

PROJECTS

WYA — What's Your Aesthetic Jan 2025 – May 2026
- Built an end-to-end AI pipeline: background removal (rembg + OpenCV) → garment masking (GrabCut) → zero-shot classification via AWS SageMaker (FashionCLIP) → KMeans colour extraction → pattern detection → rule-based fabric classifier.
- Delivered 10+ features including Style DNA Quiz, AI Outfit Matcher, Green Score, Vacation Packer, and weather-based styling.
- Architected the AWS deployment: React + TypeScript on S3 + CloudFront; FastAPI + Docker on EC2; SageMaker serverless endpoint; CI/CD via GitHub Actions; rate-limited AI endpoints; 47 pytest tests.
- Built Luna, a conversational stylist on top of the WYA API: a regex intent classifier routes structured queries to WYA endpoints, with an OpenRouter LLM fallback that receives the user's real wardrobe as context. Separate FastAPI/Docker/EC2 deployment with JWT auth shared with WYA.
- Authored a research paper on the WYA pipeline, accepted as a Full Paper at EDGEI 2026 (Springer LNNS, forthcoming).

Stack: Python · FastAPI · React · TypeScript · AWS (EC2, S3, CloudFront, SageMaker) · Docker · OpenCV · SQLite

Lumen — 32-bit x86 Operating System Present
- Built an OS from a raw boot sector up: custom BIOS bootloader and a freestanding ~13,000-line C kernel.
- Implemented paging with per-task address spaces, a bitmap frame allocator, an interrupt-safe heap, and preemptive Ring 0/Ring 3 multitasking with fork/exec.
- Added a 25-call syscall table, an ATA driver, a persistent filesystem (LumenFS v2), an ELF loader, and a text-mode GUI.
- Wrote a make test harness that boots the OS headless in QEMU and fails on any self-test failure, wired into GitHub Actions.

Stack: C · x86 Assembly · NASM · QEMU · GitHub Actions

TryOn — Build, Compare, and Perfect Outfits May – Jun 2026
- Built a computer vision pipeline that turns any clothing screenshot into a transparent garment cutout (rembg), auto-categorizes it with a FashionCLIP zero-shot classifier, and fits it onto a 2D paper-doll or rigged 3D mannequin using MediaPipe BlazePose landmarks and thin-plate-spline warping.
- Built the 3D Mannequin Garment Viewer: a CV analyzer detects strap type and back construction from one photo, with a 75% confidence gate that reports "undetermined" rather than hallucinating a back. Falls back to a client-side Canvas engine when the backend is offline.
- Tested the FastAPI backend with 69 tests at 86% coverage, run against the real CV models instead of mocks.

Stack: Python · FastAPI · React · TypeScript · React Three Fiber · MediaPipe · rembg · SQLite

FairFaceGuard — Deepfake Detection Under Skin-Tone and Lighting Variations Present
- Designed a causal research pipeline testing whether a deepfake detector's accuracy gap across skin-tone groups comes from skin tone itself or from correlated illumination cues.
- Built illuminant-corrected ITA/Fitzpatrick labelling and matched counterfactuals (skin-only, light-only, both) with an automated manipulation check.
- Interrogated a frozen EfficientNet-B4 baseline via adversarial GRL sweeps, linear probing, and a counterfactual causal test, with bootstrap CIs and a paired permutation test. Pilot run on a FaceForensics++ subset.
- Paper accepted for presentation and publication at IMPACT 2026 (Springer ICT series, forthcoming).

Stack: Python · PyTorch · EfficientNet-B4 · FaceForensics++

PUBLICATIONS

- *WYA — From Pixels to Wardrobe: A Deep Learning and Computer Vision Pipeline for Automated Garment Analysis.* Accepted as a Full Paper, EDGEI 2026 (SIUC, Malaysia); Springer Lecture Notes in Networks and Systems (LNNS), forthcoming.
- *FairFaceGuard: An Observational Fairness Audit and a Negative Result on Counterfactual Skin-Tone Control in Deepfake Detection.* Accepted, IMPACT 2026 (International Conference on Intelligent Machine Processes and AI-Centric Technologies); Springer Information and Communication Technology (ICT) series, forthcoming.

SKILLS & TECHNOLOGIES

Languages: Python · TypeScript · JavaScript · Java · C · SQL (MySQL)
Frameworks & Libraries: FastAPI · Express.js · React · Node.js · OpenCV · MediaPipe · scikit-learn · PyTorch · Pandas · NumPy
Cloud & DevOps: AWS (EC2, S3, CloudFront, SageMaker, IAM) · Docker · GitHub Actions · CI/CD
Data & ML: Data Analysis · KMeans Clustering · Computer Vision · GNNs · Zero-shot Classification · Regression
Databases: SQLite · MySQL
Tools: Git · Pytest · QEMU · Vite · Tableau · Excel

CERTIFICATIONS

- IBM Data Analyst Professional Certificate
- Google Data Analytics Professional Certificate
- Google IT Automation with Python
- Meta Data Analyst with GenAI
- Programming in Java — NPTEL

LANGUAGES

English (Native) · Hindi (Advanced) · Malayalam (Beginner)